

IMPORTANT: Title Change – “Aharonov-Bohm Effect” should be changed to “Foundational Physics”

Foundational Physics

I highly recommend reading Linear Algebra before this for a better intuition * this line should be pink*

Imagine you're outside and you pick up a rock and drop it from about the height of your head. It makes a pretty loud sound so intuitively it must have hit the ground pretty forcefully. Now you pick that same rock up and drop it from waist height, it yields a softer sound, so intuitively it hits the ground less forcefully.

Let's look at the relationship between dropping rocks from varying heights and the force *force should be red* that it gains as a result. Creating logical relationships is best done in math, so let's use math.

We have a rock which is an object that has mass, so let's call the rock M * M should be italicized* and the distance we are dropping it from D * D should be italicized*. We know that heavier objects produce more force when dropped than lighter objects, so we can associate force and mass as being proportional / equal – if an object is bigger, it can produce more force – so right now we have:

$$F = M$$

And we also know that if we drop our rock from greater heights, it accumulates greater force, therefore D must also be proportional to F as a factor with M .

$$F = M * D$$

With this relationship of vertical force, let's look at how it changes horizontally and see if we can combine the two to make a general formula.

HORIZONTAL FORCE * subheading font *

Let's take our rock again and throw it at a wall. After throwing a couple of rocks with varying sizes – or masses / M values – we can observe that bigger rocks make louder sounds when they hit the wall.

So once again we know that force produced horizontally is proportional to mass.

$$F = M$$

However you realize the amount of speed the rock has on it also affects the force produced by the rock. The faster the rock is moving, the more forcefully it hits the wall. Additionally, it's not just about how hard you throw the rock, but it's where you throw it as well.

After throwing the rock at many places on the wall, you realized the time the rock takes to hit the center of the wall is shorter than anywhere else, and your arm when it's fully extended is aligned with the center of the wall – so the rock has a straight trajectory or path of motion –

Additionally, this means the amount of force produced by the rock is greatest in the center of the wall. Because we know speed is proportional to force generated, and at the point of contact between the rock and the wall in the center, the speed the rock has on it is greatest, therefore the force it carries on it must also be the greatest. This means the force is proportional to the speed in specific directions. This means we're looking at a vector *vector should be yellow* – it has magnitude (speed) and direction – and we'll call it velocity *velocity should be yellow* because the Latin root for “speed” is *velocitas* *velocitas should be italicized*

So now we have:

$$F = M * v$$

Where v is the velocity vector.

But if this is true and the amount of force changes based on what direction we throw the rock in at the wall, then that means Force must also be a vector because it has magnitude and directional components.

This equation looks like it makes sense but there's something we forgot to consider. When we throw our rocks at the wall horizontally, the closer the wall is, the more velocity our rock retains – mathematically we can say it has a greater slope (rate of change) – until it hits the wall.

So imagine our wall is a fixed distance away, like 10 feet. We throw the rock really hard – force is measured in Newtons so we'll say x Newtons –

So our initial force will be (x Newton's) times the amount of mass the rock has, but then as it travels through the air it loses speed. This means our force vector is going to decrease in magnitude gradually over time until it hits the wall. To mathematically account for this gradual change over time, let's look at the rate of change of our velocity as time progresses.

The rate of change is useful because our rock isn't going to slow down and then speed up again mid air, it will just keep getting slower due to air resistance, therefore it has a constantly decreasing rate of change / slope. So because this value is constant, stable, and accounts for the changes in velocity over time, that's what we'll use in our force equation.

Mathematically, accounting for this gradual change of slope over time is called differentiating the velocity vector, and let's say that yields to a vector called the acceleration *acceleration should be yellow* of the rock because

we like it when our rock gets faster and in Latin, the word “accelerat”
accelerat should be italicized is used to describe that.

Ultimately, our final equation for horizontal force is:

$$F = M * a$$

Where a is the acceleration vector.

HORIZONTAL v. VERTICAL? *subheading font*

So why is Vertical Force equal to the mass of our rock times the distance from the ground and Horizontal Force equal to the mass of our rock times its acceleration vector?

Well they actually aren't different.

Let's say they are equal and see what this has to imply mathematically by setting the equations to each other.

$$M * d = M * a$$

– Divide by m on both sides –

$$D = a$$

Distance equals acceleration. This sounds absurd but let's break it down.

When we drop our rock from a height, it picks up speed straight down. Now instead of the rock getting slower over time because of air resistance, it actually gets faster. So knowing that, we can say greater falling time leads to greater velocity, so this means a constant increasing slope for our velocity vector – which is a positive acceleration vector – due to the increasing velocity.

Falling time is directly proportional to distance in a situation where we keep other variables constant. An object that has to fall 100 meters will obviously take more time than an object which has to fall 10 meters, traveling through the same conditions at the same initial speeds. However, it's a proportional situation because if it was linear then that means falling 100 meters would take exactly 10 times longer than falling 10 meters.

Because our velocity vector has a slope which is constantly increasing, the speed keeps increasing, therefore the object falling for longer picks up more speed than the other object falling for less time, and therefore it becomes a non-linear situation.

So we know

Falling Time = Velocity,

Let's differentiate – or look at how this relationship is changing more fundamentally – and see what happens.

We know falling time is proportional to distance, but to be exact, it is proportional to distance with respect to time. An object's falling time literally measures the amount of time it takes an object to travel a certain distance.

Mathematically we denote that as:

Distance = d/dt (Falling time)

And we already know that acceleration is measuring how velocity changes over time,

$a = d/dt$ (Velocity)

Setting distance equal to acceleration means

d/dt (Falling time) = d/dt (Velocity)

And that is the same as differentiating our original equation which we know is true, with respect to time:

$$d/dt (\text{Falling time} = \text{Velocity})$$